

Weekly Learning Debrief

MIS410 — Data Analytics Certificate | Week 6

Topics: NumPy fundamentals, statistical anomaly detection (Z-score, IQR), machine learning anomaly detection (Isolation Forest)

Reading: NumPy Fundamentals

Covered NumPy's core structure and speed advantage — the foundation underneath pandas that's been running quietly since week one.

- **ndarray:** NumPy's core data structure — every array result (`np.arange`, `np.random.standard_normal`) has been an ndarray all along.
- **Vectorization:** doing math on an entire array at once instead of looping item by item. Proven with `%timeit`: array multiplication ran ~68x faster than an equivalent Python list loop.
- **Broadcasting:** NumPy automatically applies an operation across an array without manually matching every element (e.g. `data * 10` doubles every value in a 2D grid in one line).
- **Multi-dimensional indexing:** 3D arrays as “stacked trays” — `arr3d[0]` grabs a whole layer, `arr3d[1, 0]` drills into a specific row within a specific layer. A single index flattens a dimension; a colon-range preserves it.
- **isnan:** NumPy's raw-array-level cousin of pandas' `.isnull()` — same missing-value detection, different layer.
- Random walk example: simulating a coin-flip-driven position over 1,000 steps — directly relevant to anomaly detection, since it models how a metric can drift unpredictably over time, and “normal randomness” vs. a genuine anomaly is exactly what this week's assignment distinguishes.
- **savez / savez_compressed:** NumPy's own save/load format for arrays — the `.npz` equivalent of pandas' `.to_csv()`.

Assignment: SQL + Python Anomaly Detection

Built a full anomaly detection pipeline on transaction data — SQL retrieval/aggregation, then three independent Python detection methods compared against each other.

Method	On transaction_amount	On merchant transaction_count
Z-score (± 3)	0 anomalies	Some merchants flagged
Z-score (± 2 , loosened)	0 anomalies	—
IQR ($Q1/Q3 \pm 1.5 \times IQR$)	0 anomalies	Some merchants flagged (partial overlap with Z-score)
Isolation Forest (ML)	5 anomalies found, near range extremes	—

- **Key finding:** Z-score and IQR agreed — zero anomalies in transaction amounts, consistent with a roughly uniform histogram (no long tail). Isolation Forest, working on a completely different principle (ease of isolation rather than distance from mean), found 5 anomalies near the extreme edges of the range that both statistical methods missed.
- **Customer time series:** daily transaction count almost always exactly 1/customer/day, with 4 visually notable 2-transaction days — correctly distinguished as an observation, not a statistically confirmed anomaly, since no formal method flagged them.
- **Merchant-level cross-method comparison:** built one combined table showing which merchants were flagged by both Z-score and IQR (strong, corroborated anomalies) vs. only one method (borderline cases) — box plot visually confirmed 2 outlier merchants at 4 and 5 transactions.
- **Corrected in the write-up:** an early draft conflated the loosened ± 2 Z-score threshold with IQR's own separate quartile-based formula — fixed to accurately describe two independent methods, not one method described two different ways.

Discussion Posts

- Main post used a movie-industry example: a “cult classic” framed literally as a data anomaly — a film whose performance sits far outside the expected cluster predicted by cast, genre, and production factors.
- Reply discussed the adversarial angle — as ML-based fraud detection (like Isolation Forest) becomes more common, can bad actors adapt to evade it, similar to how phishing tactics evolved past older spam filters.

Concept Worth Remembering

- **Different anomaly detection methods define “unusual” differently.** Z-score/IQR measure distance from center; Isolation Forest measures ease of separation from the crowd. Agreement across methods is a stronger signal than any single method alone — and a “zero anomalies found” result is a legitimate, reportable finding, not a failed analysis.

Next steps: dedicated standard deviation deep-dive session planned after this week wraps.